

Building a universal game-playing agent infrastructure to optimize performance

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IM Topic presentation

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LLMs as agents

- Models predict, agents reason, act and interact
- Agents possess **agency**: the ability to perform controlled actions in their own will
- A language agent carries out goal-directed behaviour, while still being just a next token-predictor

[Plaat, Duijn et al. 2025]

Game-playing agents

- Game goals (e.g., guess the word in Wordle or win a chess match) are **ill-posed** [Horst, Mazzaccara et al.2025]

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Math problem (well-posed)	Dialogue games (ill-posed)
- One right answer - One (or a finite set of) way(s) to solve	- One or many right answers - A large number of ways to win (which word to start with in Wordle / different chess strategies / collaborating strategies)

- Generally, Dialogue games also possess a larger context size (cf. the amount of information the Player has to “remember” in Taboo and the context of solving a math problem (even a large one with brackets etc.)
- The context is complex: game master comments, current game state, remembering the game rules etc.

Research problem and some caveats

Goal: create a general dialogue game-playing agent by adding different components around vanilla LLMs

Important side-notes:

- Clembench is a large collection of games (collaborative, word-guessing, environment interaction, multimodal) – improvements need to be on all of them
- Can't build specialized solver
- The clembench game flow can't be violated

Improvement methods

Three paths of improvement:

- 1) Adding components (Reflection, ReAct)
- 2) Adjusting the prompting methods (Few-Shot, Tree of Thoughts)
- 3) External tools (databases)

Previous articles on the topic combine them – these are just points of variation

Improvement methods: external evaluation

Reflect [Shinn, Cassano et al. 2023] — using verbal reinforcement to teach agents to learn from past mistakes

- A more enhanced version of setting the context to the player after the ParseError / other error
- These reflections are stored in the “memory” and used in future trials to improve decision-making
- The authors report a success rate improvement in the decision-making game AlfWorld by 22%
- If reflecting after each round, it might help the model not to get stuck repeating the same answer again and again

Improvement methods: external evaluation

Welcome to Wordle! You have six attempts to guess the target word, a valid English word of five lowercase letters (a-z). Please use the tags <...>

CLUE: mug

The answer is wrong. Please try again

```
perspective = [  
  { "role": "user",  
    "content": "Welcome to Wordle! <...>." },  
  { "role": "assistant",  
    "content": "CLUE: mug" },  
  { "role": "user",  
    "content": "The answer is wrong. Please try again" } ]
```

Updated perspective of the player

I failed to follow the Wordle rules. The first guess, “mug”, contained only three letters, even though the instructions <...>.

Reflection model

Improvement methods: external evaluation

Reflect [Shinn, Cassano et al. 2023] — using verbal reinforcement to teach agents to learn from past mistakes

- Reflecting model's choice: same or separate?
 - If the same model is reflecting, the reflection setup is almost equal to the classic clembench setup
 - If allowed, use a pre-trained model
-
- False negatives: a good player's move may be reflected in a wrong way

Improvement methods: external evaluation

ReAct [Yao, Zhao et al. 2023] – synergized thinking and doing

- A task from the GM $\Rightarrow$ a thought $\Rightarrow$ an action
- Another player's move (or response from the environment) serves as an observation
- An observation induces as a new task $\Rightarrow$ thought $\Rightarrow$ action $\Rightarrow$...

Improvement methods: external evaluation

GM

<...> Taboo <...>
Target word: flashlight
Related words: flash, light

Player 1



- thinking component ->

I need to think where the flashlight is used.
Does it often occur together with some
words that I'm not forbidden to use? <...>

Improvement methods: external evaluation

GM

<...> Taboo <...>
Target word: flashlight
Related words: flash, light

The other player's guess was torch

Player 1

CLUE: <...>

thinking component

I need to think where the flashlight is used.
Does it often occur together with some
words that I'm not forbidden to use? <...>

Improvement methods: external evaluation

GM

<...> Taboo <...>

Target word: flashlight
Related words: flash, light

The other player's guess was torch

Player 1

CLUE: <...>

thinking component

I need to think where the flashlight is used.
Does it often occur together with some
words that I'm not forbidden to use? <...>

thinking component

I probably did something wrong. <...>

Improvement methods: external evaluation

ReAct [Yao, Zhao et al. 2023] – synergized thinking and doing

- Complex tasks with large action spaces will easily exceed the context window
- Output length has to be controlled

Improvement methods: internal evaluation

Prompt adjustments based on [Lin, Roberts et al. 2025]

- Instead of telling the model to just think, clear intermediate steps are provided
- An additional role setting (“You’re an expert playing a game...”)

Improvement methods: internal evaluation

GM

Player

You are an expert playing the AdventureGame.

[RULES]

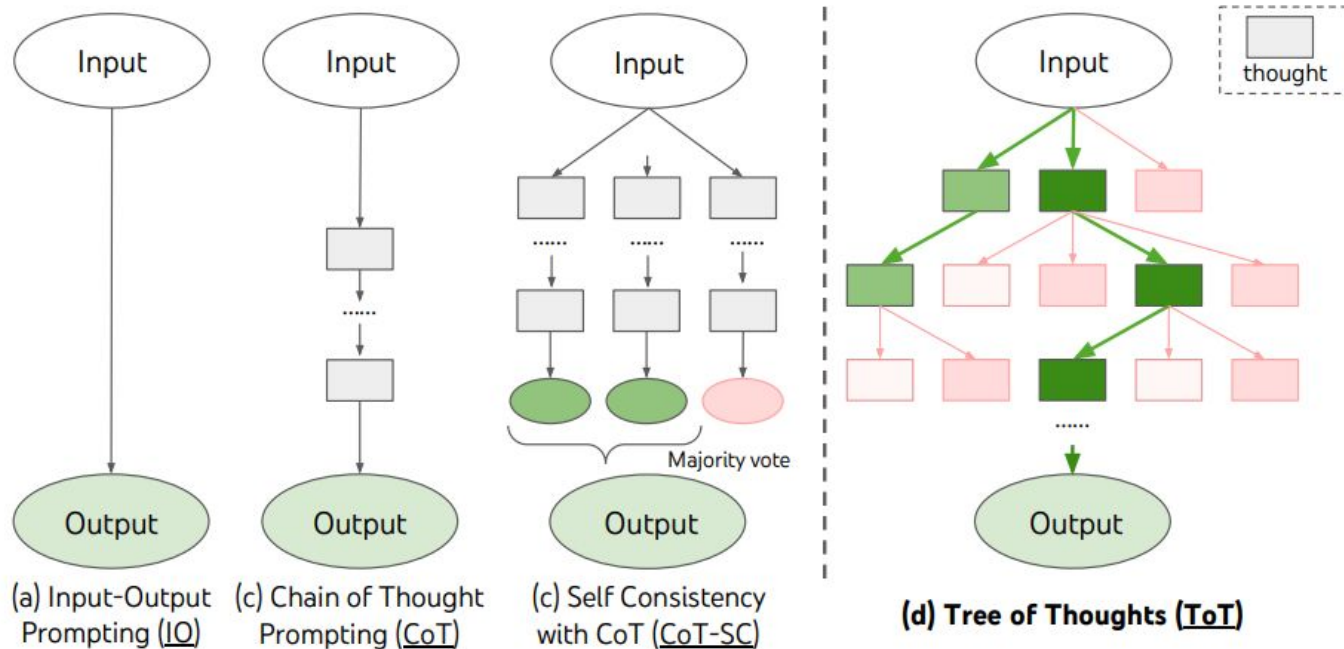
Before making a decision, articulate your internal thinking process. Your performance will be assessed on both the intermediate thinking results and the final decision. Follow the thinking process:

1. **Current Position Analysis** * State the coordinates of your current position <...>.
2. **Valid Actions** List all possible actions <...>
3. **Strategic Analysis** * Explain your reasoning for choosing the final action, considering factors like: <...>
<...>
5. **Chosen Action** * In this section, write the single action you want to take in the game starting with >
For example: <...>

>take plate

<...>

Improvement methods: CoT and ToT



[Yao, Yu et al. 2023]

Improvement methods: CoT and ToT

- Generates reasoning trajectories
- CoT alone does not give any promising results
- Whereas ToT has a high computational and implementation cost
- Under ToT, hallucinations may be amplified

Methodology

- Baseline: vanilla LLMs' gameplay results
- Compare improvements across open-weight vs. closed-weight models
- Evaluate different categories of games separately (grouped by gameplay time or abilities tested)
- Test the above mentioned methods both individually and in combination
- Incorporation of general-domain datasets (e.g., QA or spatial reasoning) possible

Project timeline

November:

- clembench games exploration (especially multimodal)
- finalizing possible improvement paths (and their combinations)

December:

- improvement paths implementation, testing
- refactoring and testing other paths if no improvements

January:

- improvement must be achieved by this point
- additional testing on general-domain datasets
- visualizing the results
- ablation studies

February:

- writing the report
- preparing the poster

References

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